

THE WARM LINE

京张暖线

Giving AI's waste heat, its compute and its off-switch back to the city

Centennial Jing-Zhang AI Innovation Belt open call · concept proposal · A3 booklet

zxw1992 / jingzhang-warm-line · generated by Claude Opus 5

Concept suggestion · not statutory planning · no government decision · provisional base boundary

AI does two very physical things to a city: it makes heat, and it needs to be near. This proposal turns both into spatial rules — waste heat is returned to public space, compute comes down to street level, and the switch that stops it stays in human hands.

Proposal term	Jing-Zhang prototype	Spatial content
Compute Watering Station	Water station	Edge compute + charging + public waste heat + services
Dispatch Hall	Dispatch office	A public task queue with a human stop lever
Marshalling Yard	Marshalling yard	Assembling, running and reviewing test scenarios
Stitching Link	Level crossing	Pedestrian-priority link and scenario interface
Milestone	Mileage marker	Honour display and contributor record along the line

Core metrics (recomputed in EPSG:4548)

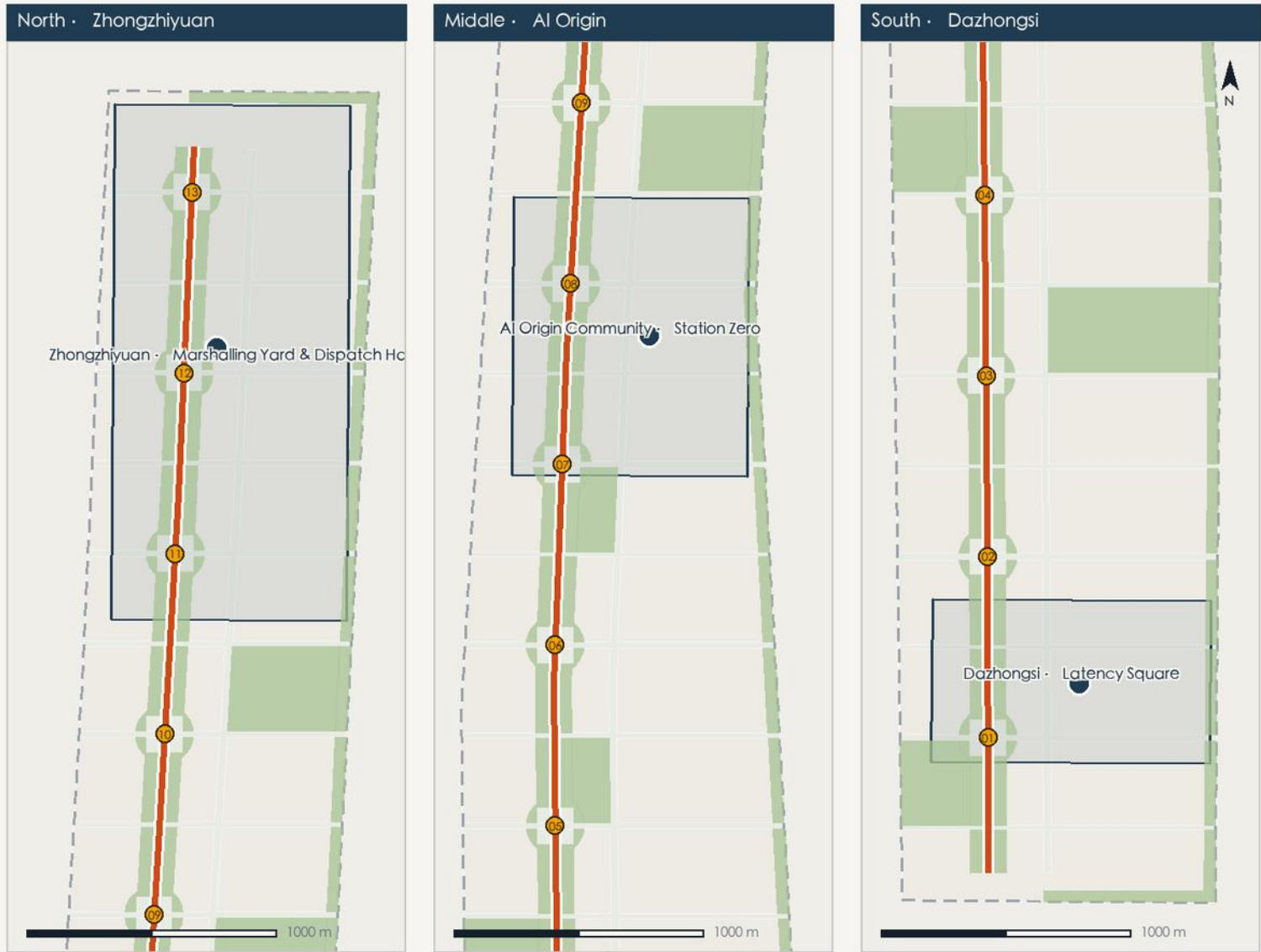
Overall design area 11.41 km ²	Green ratio 19.7 %	Public space 5.9 %	Road land 10.2 %	Stations 13 no.
Stitching links 13 no.	Spine length 9.39 km	Warm arcade 33.1 ha	Demo coverage 17.0 %	Demo bulk check 1.87

Fig.1 Overall spatial structure: THE WARM LINE

One spine · 13 compute watering stations · 13 east-west stitching links · three areas, two wings

SITE OVERVIEW

Concept proposal · not statutory planning · provisional boundary
zxw1992 / jingzhang-warm-line



Design position

AI does two very physical things to a city: it makes heat, and it needs to be near.
This proposal turns both into spatial rules —
waste heat is returned to public space, compute comes down to street level, and the switch that stops it stays in human hands.

Two wings



Three scope levels

43.6 km ²	Coordinated research area
11.4 km ²	Overall design area
368.4 ha	Key detailed-design areas

Legend

- Warm Line spine (heritage-park slow line)
- Compute Watering Station (edge compute + public waste heat)
- AI pilgrimage landmark
- Park green / buffer green
- Key area (provisional extent)
- Overall design area (provisional boundary)

Naming system: from steam-era resupply to agent-era resupply

Proposal term	Jing-Zhang prototype	Spatial content
Compute Watering Station	Water station	Edge compute + charging + public waste heat + services
Dispatch Hall	Dispatch office	A public task queue with a human stop lever
Marshalling Yard	Marshalling yard	Assembling, running and reviewing test scenarios
Stitching Link	Level crossing	Pedestrian-priority east-west link and scenario interface
Milestone	Mileage marker	Honour display and contributor record along the line

The base is a provisional rough boundary inferred from the announcement's written extents (provisional constraint, official_boundary=false). It supports concept generation, display and self-check only; it is not an official redline, an approval basis or a precise-area basis. All spatial ideas are concept suggestions for professional teams to develop. Areas are recomputed in EPSG:4548; see sources.json.

Fig.2 Land-use structure and the three scope levels

Concept land-use structure over the provisional boundary (not a statutory land-use plan)

LAND USE

Concept proposal · not statutory planning · provisional boundary
zxw1992 / jingzhang-warm-line



Structure logic: one partition built from four subtractions

- 1 Road land**

Streets and stitching links are subtracted first to give a readable block skeleton
- 2 Station squares**

Forecourts are subtracted at the 13 links, so public space precedes buildings
- 3 Green spine**

The continuous park and station gardens follow the heritage slow line
- 4 Buffer green**

A protective strip is subtracted along the eastern traffic edge
- 5 Remaining blocks**

What is left is coded by the three-areas-two-wings logic, completing the partition

Codes follow the project subset of the national land-use classification guide. This is a concept structure, not a regulatory boundary, ownership conclusion or development-intensity conclusion.

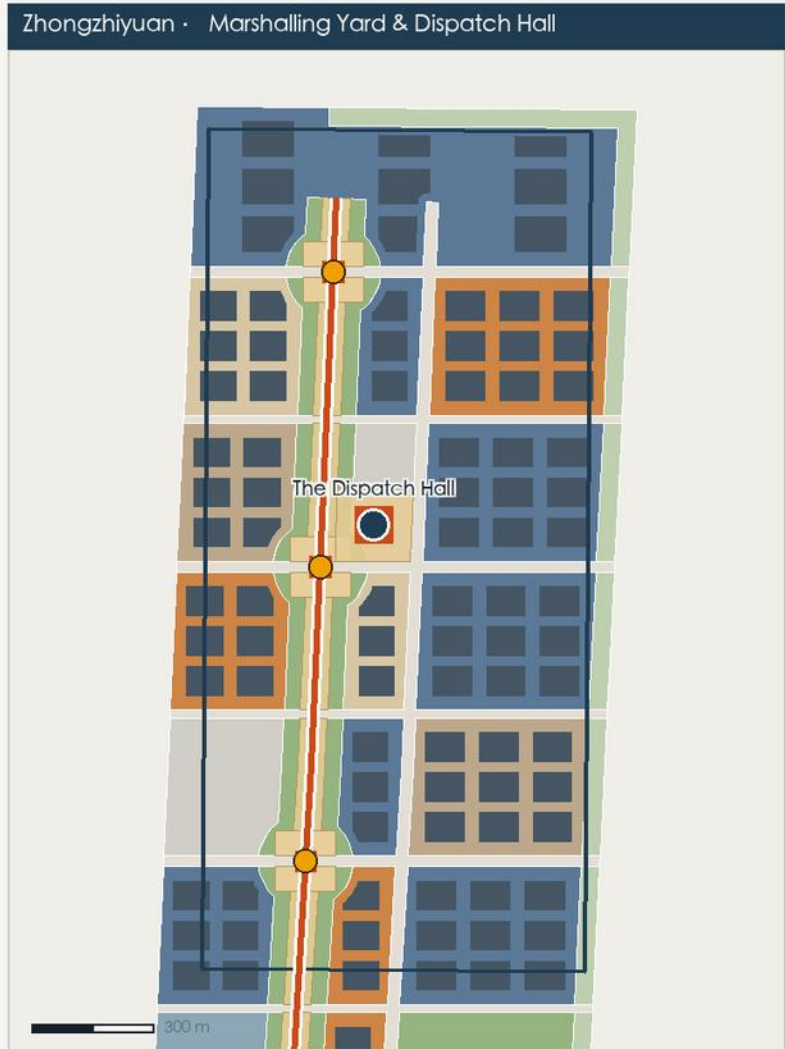
The base is a provisional rough boundary inferred from the announcement's written extents (provisional constraint, official_boundary=false). It supports concept generation, display and self-check only; it is not an official redline, an approval basis or a precise-area basis. All spatial ideas are concept suggestions for professional teams to develop. Areas are recomputed in EPSG:4548; see sources.json.

Fig.3 Detailed design of the three key areas

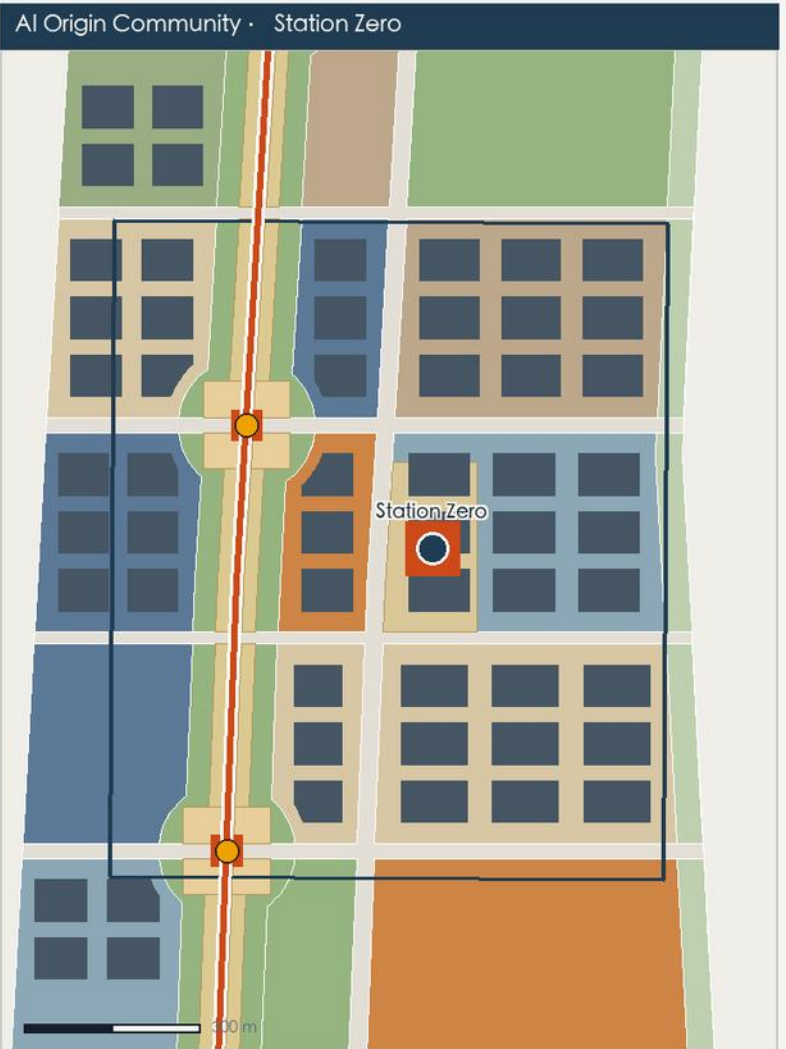
Zhongzhiyuan / AI Origin Community / Dazhongsi: programme, public space and landmarks

KEY AREAS

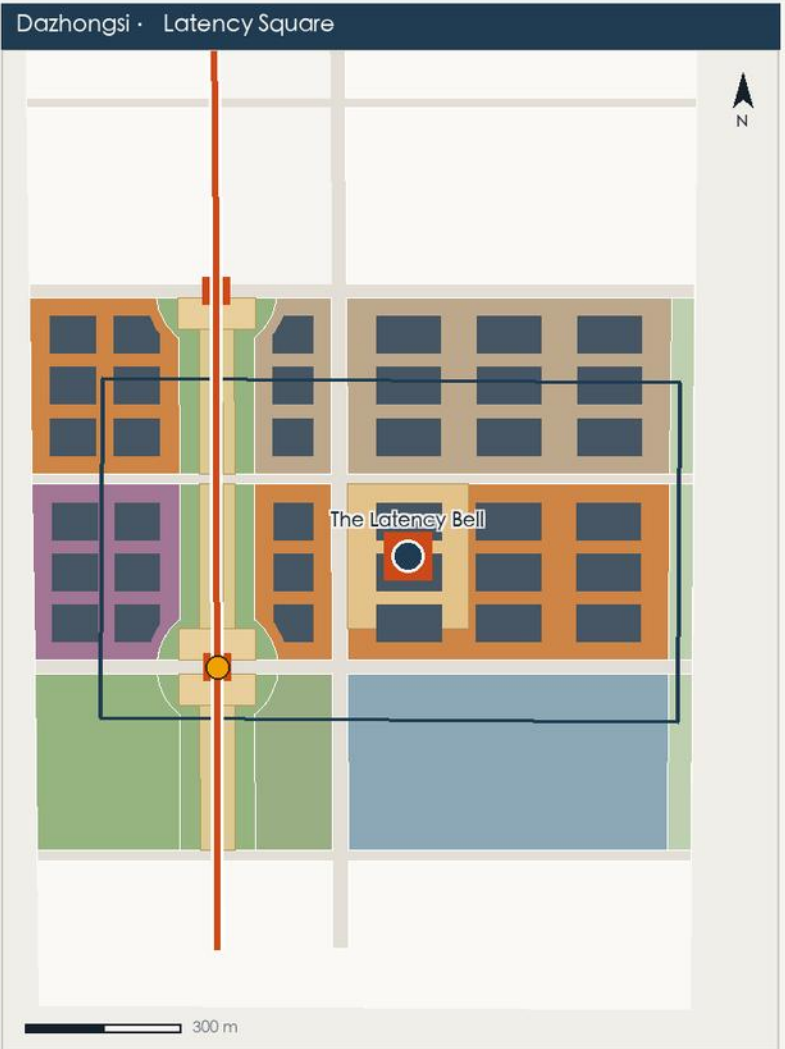
Concept proposal · not statutory planning · provisional boundary
zxw1992 / jingzhang-warm-line



- 192.1 ha · north
- Marshalling yard: industry test scenarios assembled, run and reviewed
 - Dispatch Hall: a public task queue with a human stop lever
 - Reserved land keeps compute and heat infrastructure options open



- 104.3 ha · middle
- Station Zero: server hall, waste-heat bathhouse and open-source commons in one building
 - Fine-grained blocks provide affordable space to fail and retry
 - Origin square hosts releases, post-mortems and public dialogue



- 72.0 ha · south
- Latency Square: from bell-tower timekeeping to millisecond dispatch
 - Intelligence-native retail and business around the transit station
 - Southern start of the milestone and honour-display system

■ Concept building footprint ■ Stations and landmarks ■ Public space (squares & arcade) ● Compute Watering Station (edge compute + public water area) (provisional extent)

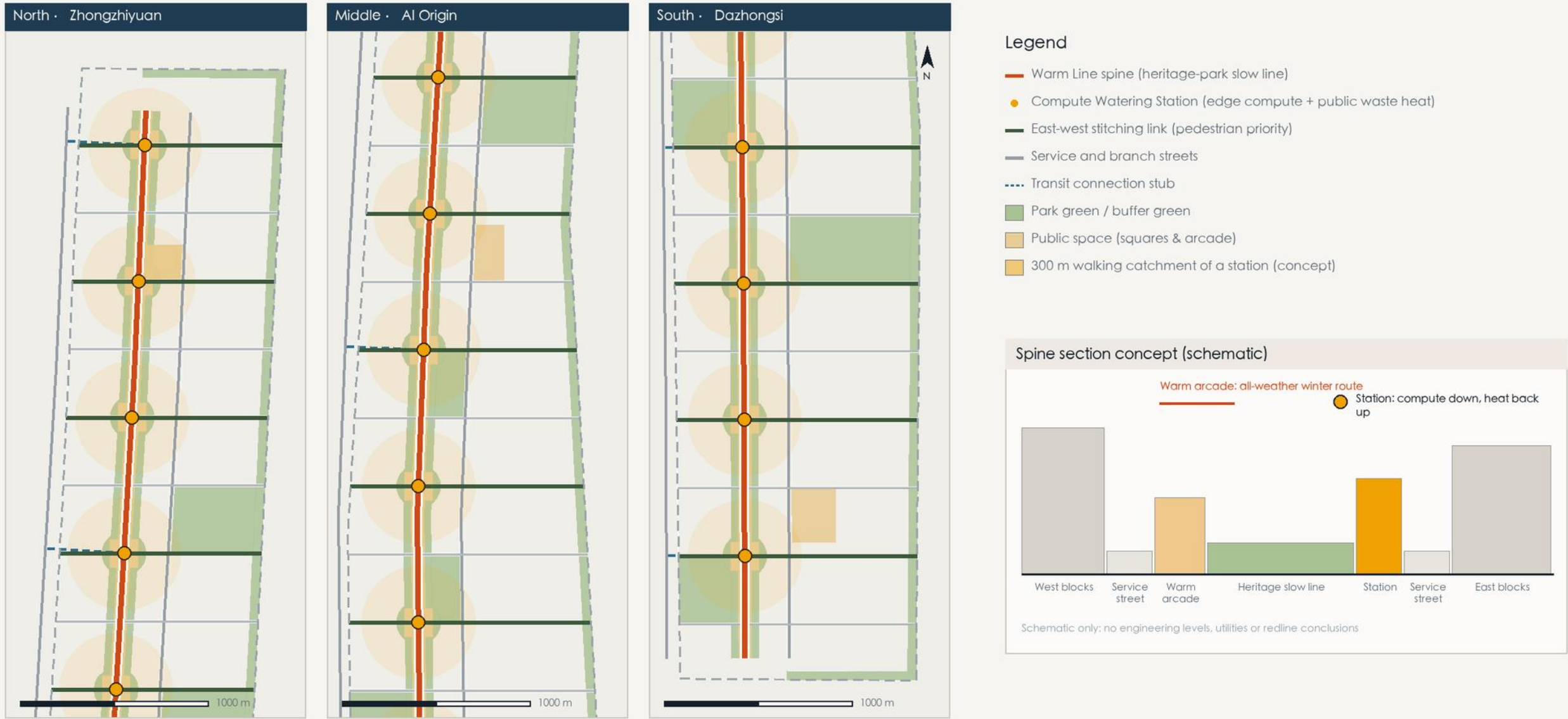
The base is a provisional rough boundary inferred from the announcement's written extents (provisional constraint, official_boundary=false). It supports concept generation, display and self-check only; it is not an official redline, an approval basis or a precise-area basis. All spatial ideas are concept suggestions for professional teams to develop. Areas are recomputed in EPSG:4548; see sources.json.

Fig.4 Mobility, blue-green network and public space

A continuous system of slow line, warm arcade, stations and stitching links

MOBILITY & BLUE-GREEN

Concept proposal · not statutory planning · provisional boundary
zxw1992 / jingzhang-warm-line



System takeaways

- The spine is 9.39 km long; 13 stations sit about 722 m apart, so the 300 m walking catchments stay continuous.
- Green ratio 19.7%, public-space ratio 5.9%, including a 33.1 ha warm arcade for all-weather winter movement.
- 13 pedestrian-priority stitching links reconnect the two sides that the railway separated for a century; each link is also a scenario interface and a human stop point.
- Transit stubs, low-speed delivery and robot resupply share the stations instead of each claiming street space.

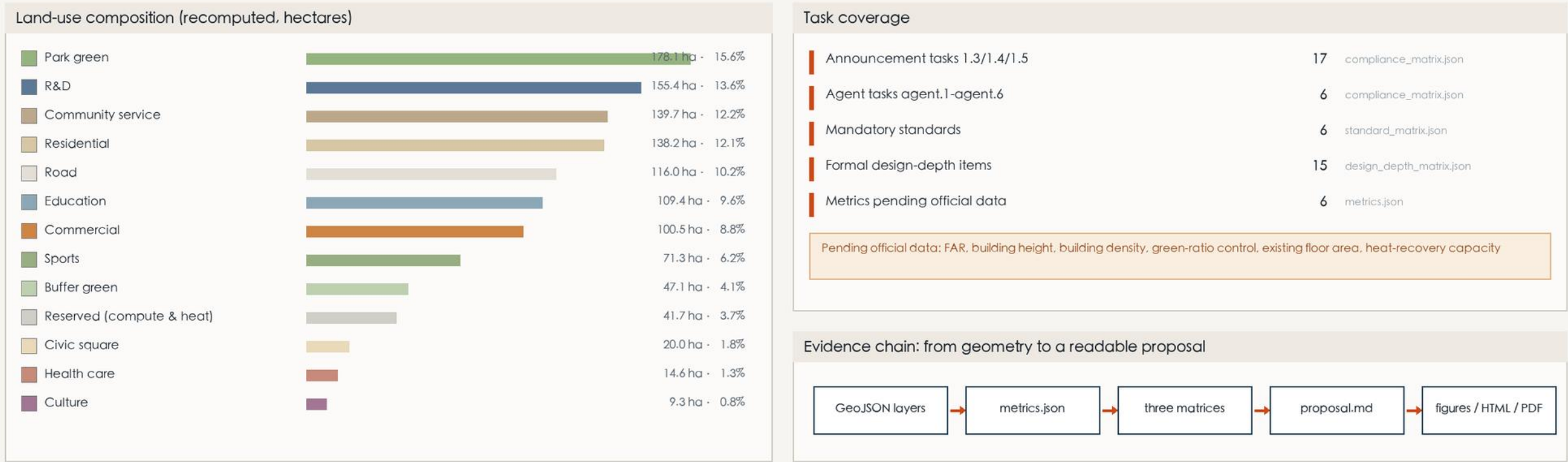
The base is a provisional rough boundary inferred from the announcement's written extents (provisional constraint, official_boundary=false). It supports concept generation, display and self-check only; it is not an official redline, an approval basis or a precise-area basis. All spatial ideas are concept suggestions for professional teams to develop. Areas are recomputed in EPSG:4548; see sources.json.

Fig.5 Metric recomputation and evidence chain

Every number is recomputable from the GeoJSON layers in EPSG:4548

METRICS & EVIDENCE

Concept proposal · not statutory planning · provisional boundary
zxw1992 / jingzhang-warm-line



Recomputation method and limits

All areas are recomputed in EPSG:4548 (CGCS2000 3-degree zone, CM 117E); the GeoJSON exchange CRS is EPSG:4326. The land-use layer is a complete partition of the submitted boundary: adjacent polygons share identical edge coordinates, their union equals the site area with a gap below 0.001%, and no polygons overlap. Every spatial layer is built on the provisional rough boundary. FAR, building height, building density, green-ratio control, existing floor area and heat-recovery capacity stay marked as pending official data. This package cannot support statutory planning, engineering feasibility or implementation commitments; once official polygons are published the whole chain must be recomputed and all drawings, HTML and metrics reissued.

The base is a provisional rough boundary inferred from the announcement's written extents (provisional constraint, official_boundary=false). It supports concept generation, display and self-check only; it is not an official redline, an approval basis or a precise-area basis. All spatial ideas are concept suggestions for professional teams to develop. Areas are recomputed in EPSG:4548; see sources.json.

Twelve AI scenario cards (summary)

- S01 Low-speed delivery and robot resupply
- S03 Waste-heat dispatch and winter arcade
- S05 Community health navigation
- S07 Public safety and event review
- S09 Station energy and carbon dashboard
- S11 Civic task queue and interruption
- S02 Walking gap and accessibility audit
- S04 AI cultural guidance
- S06 Enterprise service copilot
- S08 Embodied robot outdoor testing
- S10 Open-source release and post-mortem
- S12 Latency visualisation

Long-term operation and events

- Spring · Open-Source Release Week (Origin Square)
- Summer · Outdoor Test Season (Xiaoyue River wing)
- Autumn · Civic Dispatch Assembly (Dispatch Hall)
- Winter · Warm Line Season (warm arcade)
- Brand assets: belt logo + milestone system + record of halted runs
- Conversion: visitor → developer → team → company → staying on the belt

Risks and boundary statement

- Spatial dispute: the provisional boundary may differ from the official redline; the chain is recomputable
- Technology maturity: heat upgrading and network connection start from one station demonstration
- Data privacy: no facial-recognition-driven public management; human interruption and appeal retained
- Equity: every station keeps a staffed counter and a non-digital alternative
- Six statutory metrics remain pending; this package draws no regulatory, engineering or delivery conclusion